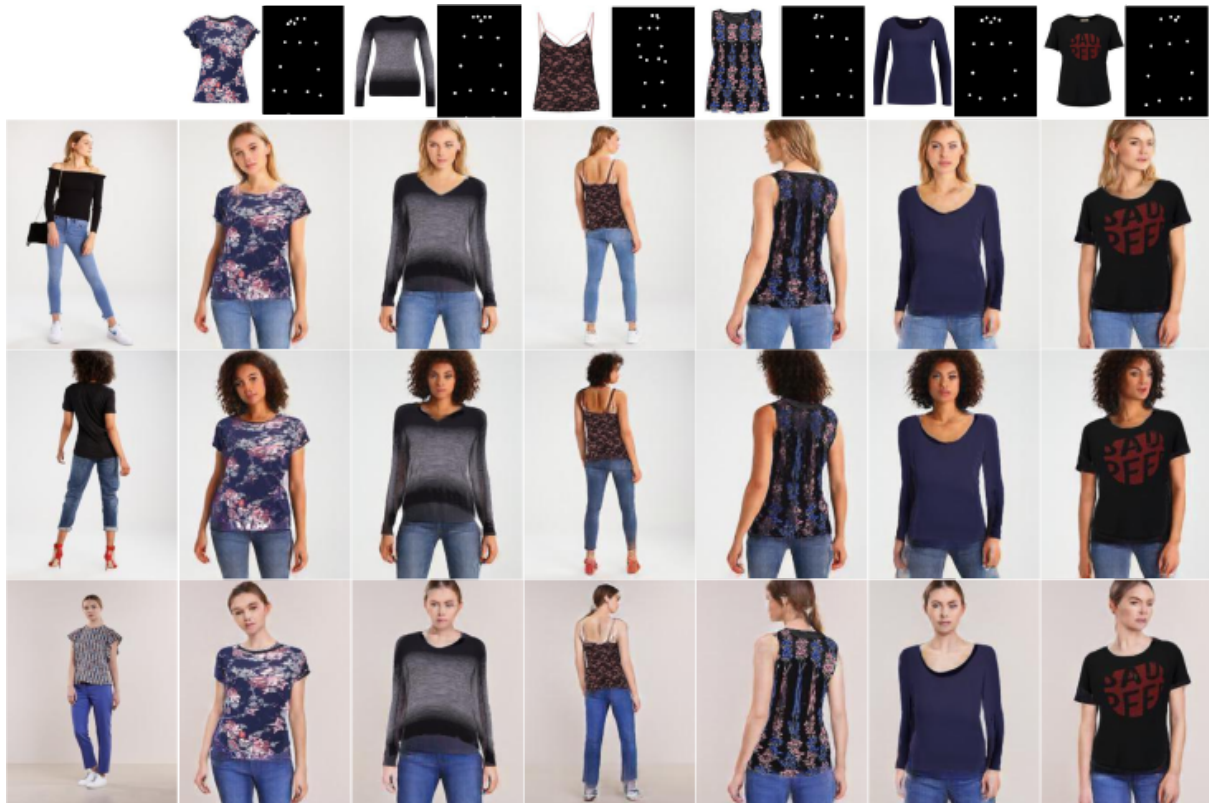

2nd year - Internship Report

Virtual Try-On Networks Study at the University of Ljubljana



The university laboratory of Ljubljana studies artificial intelligence applied to visual processing. This field grows faster than almost any other even in the computer industry. During this internship papers were published that influenced my work on several occasions. I was studying in France at home during the quarantine and soon after. For three months, my contact with the university was limited to mails and visioconference once a week. My reference was a professor named Vitomir Štruc. He helped me find sources and understand some architectures.

My goal during this internship was to reimplement some Virtual Try-On Networks and study them. I was asked to learn and report about their failures, quality measurement, dataset and updates be made. I tried to reimplement VITON and CP-VTON without success. Outside reasons detailed in the study, my computer is not able to train a network and files were written with error and incompatible modules

in a variety of languages with hardly any documentation. After browsing a tenth of paper, I finally successfully ran CP-VTON+, DeepFashion Try-On and “Down to the Last Detail : Virtual Try-On with Detail Carving” codes. Some results are shown in the study. I was also working on the history of the networks from their basic architecture to their optimized parameters.

I propose here a compilation of my founding. I hope that it will serve as an introduction for the laboratory team to their future project on VITON.

Introduction : What is a Virtual Try-On Network ?

1 - In what context, a virtual try-on is needed ?

With the emergence of the internet, traditional businesses have turned into online shops. On the one hand, online shopping is attractive for choices, prices, time-efficient, ergonomics and accessibility. On the other hand, the consumer needs to trust the piece of clothing to fit him without the ability to try it physically. Not only he can't be sure the size is right for him, he can know if it will look good on him. With the billion dollar industry that represents the only fashion market, any viable solution has a huge demand.

Various methods try to help the consumer figure this out. Pictures of models wearing the piece of clothing are usually proposed but they are limited, very expensive and can't align with every consumer body. Some papers propose to generate a 3D model of the clothing on a personalisable body but the computation time and the lack of quality results force fashion companies to search for another way. The physics of the clothing is a complex one and can't be generalised easily. Size of cloth changes. Printed texture and embroidery can be distorted. People's poses are unlimited in variety. Arms, head and other cloth need to be taken into account. At first glance a virtual try-on not only can't replace a real try-on, it looks impossible to process.

2 - What is the history of virtual try-on networks ?

Image generation is a field in intense growth since increasing computing power allowed machine learning to manage big data. Modern computer vision began its revolution in 1998. LeNet-5, Yann LeCun's Convolutional Neural Network (CNN) to recognise objects from a picture, outperform all other models. From two months 2014's CNN, Ian Goodfellow and his team train Generative Adversarial Networks (GAN) to image recognition and abstraction as well as synthesis and processing. GAN are powerful algorithms limited by data supply and computer

capacity. Every year, multiple papers keep pushing GANs boundaries with higher resolution of realistic image processing like generating faces with a list of characteristics or changing any photographie into a Munch painting.

Zalando Research released in 2017, a dataset of fashion articles in-shop pictures and wear by models. Nikolay Jetchev and Urs Bergmann, used it to train Conditional Analogy GAN (CAGAN) to swap fashion articles on people images. Two month later, Xintong Han and his team at the University of Maryland, adresse Zalando's issue with CAGAN. Named Virtual Try-On Network (VITON), their proposed architecture warps the target clothing and composes it with the GAN coarse using a CNN. VITON condition its GAN with a body mark and a pose heatmap from a pose estimator. The training and testing are performed on pictures of cloth (in-shop and on model) databases. With its architecture, VITON made better results than anything like before.

However, the composition method of VITON tends to make details transparent. About a year later, by going toward Characteristic-Preserving Image-based Virtual Try-On Network, Bochao Wang, Huabin Zheng and their team at Sun Yat-sen University made CP-VTON the current state-of-the-art architecture. Following the release of CP-VTON and up to this day, various approaches have been proposed to address its issues and limitations. It is hard to know which paper is the current best because there is no quantitative measurement of a successful Try-On.

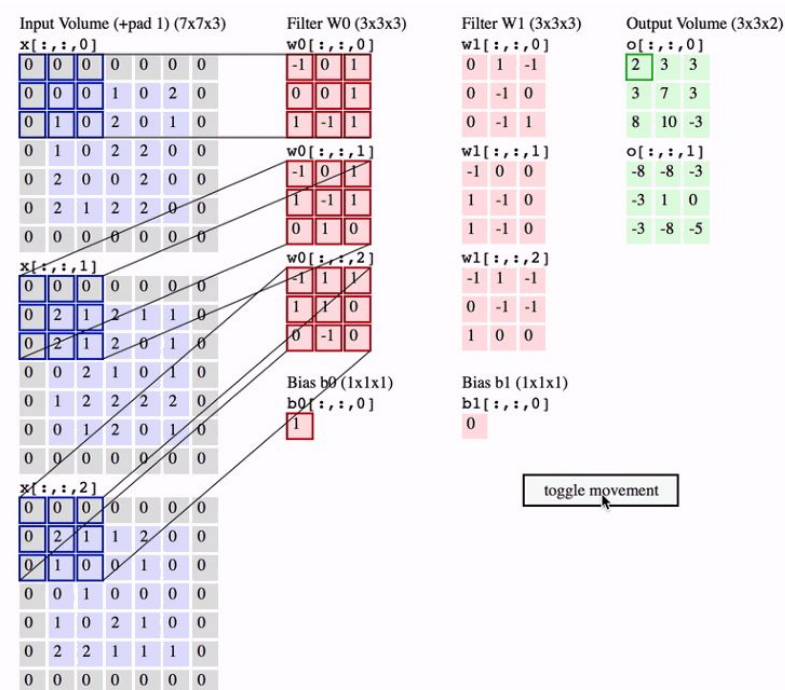
This report will study virtual try-on networks architectures, issues raised and their various ways of solving them.

I - How CP-VTON is working ?

1) What is a Convolutional Neural Network ?

An image is a high dimensional object. It can't be analyzed just, pixel by pixel. To analyse a picture, we need an algorithm invariant by translation and redundant.

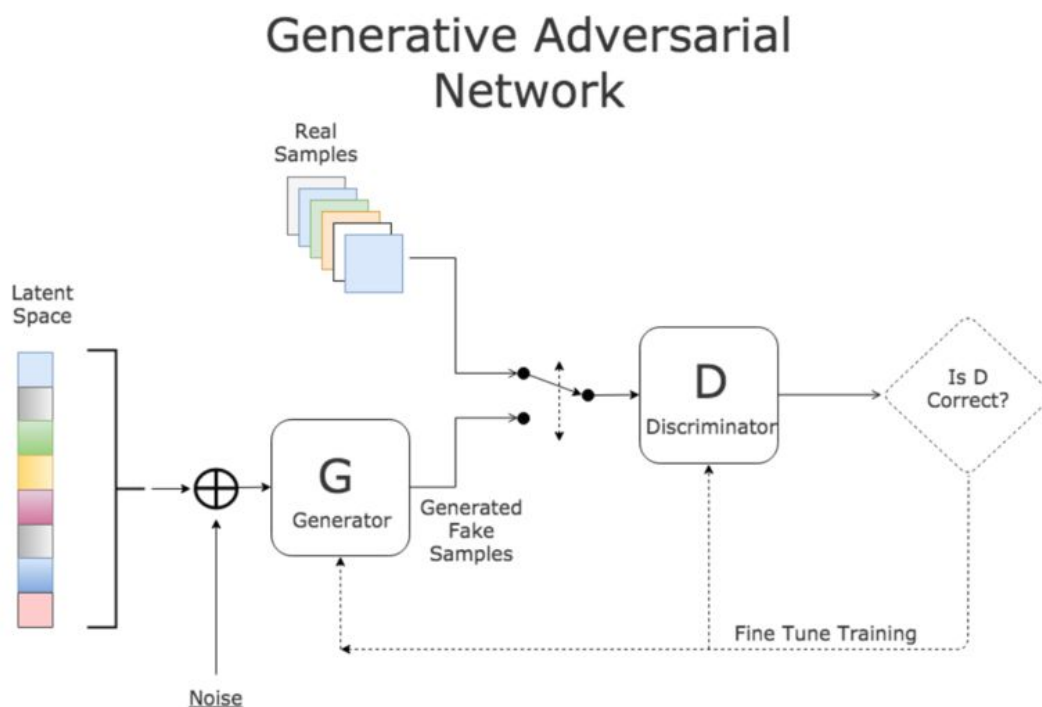
A kernel is a kind of filter which can detect specific patterns. It is a matrice of a smaller size than the image. It is generated by convolution with the image, a feature map as shown in IMAGE A. This map informs where the specific pattern was detected on the image. For nonlinear application, it is commun to reduce all non positive output to zero with a rectifier like the linear unit (ReLU).



A color picture is a matrix with a width, a length and a depth of 3, for Red, Green and Blue (RGB). To process a 3 dimensional matrix, we use a 3 dimensional kernel. Those principles can be generalized to higher dimensions. For each kernel, a feature matrix is computed. A convolution layer outputs a 3D feature map with a depth corresponding to the number of kernels used. Yann LeCun, inspired by biological visual processing, built a Convolution Neural Network from a series of convolutional layers with rectifiers and a pooling to reduce the size. Each train kernel seems to detect patterns more and more complex and detailed as the feature map gets more abstract. Kernels parameters are determined during a training session thanks to gradient descent or similar methode. The training can work only if the training data set is representative, big and wild enough and every picture labelize. The historical dataset for visual recognition is MNIST with its 70000 handwritten digits.

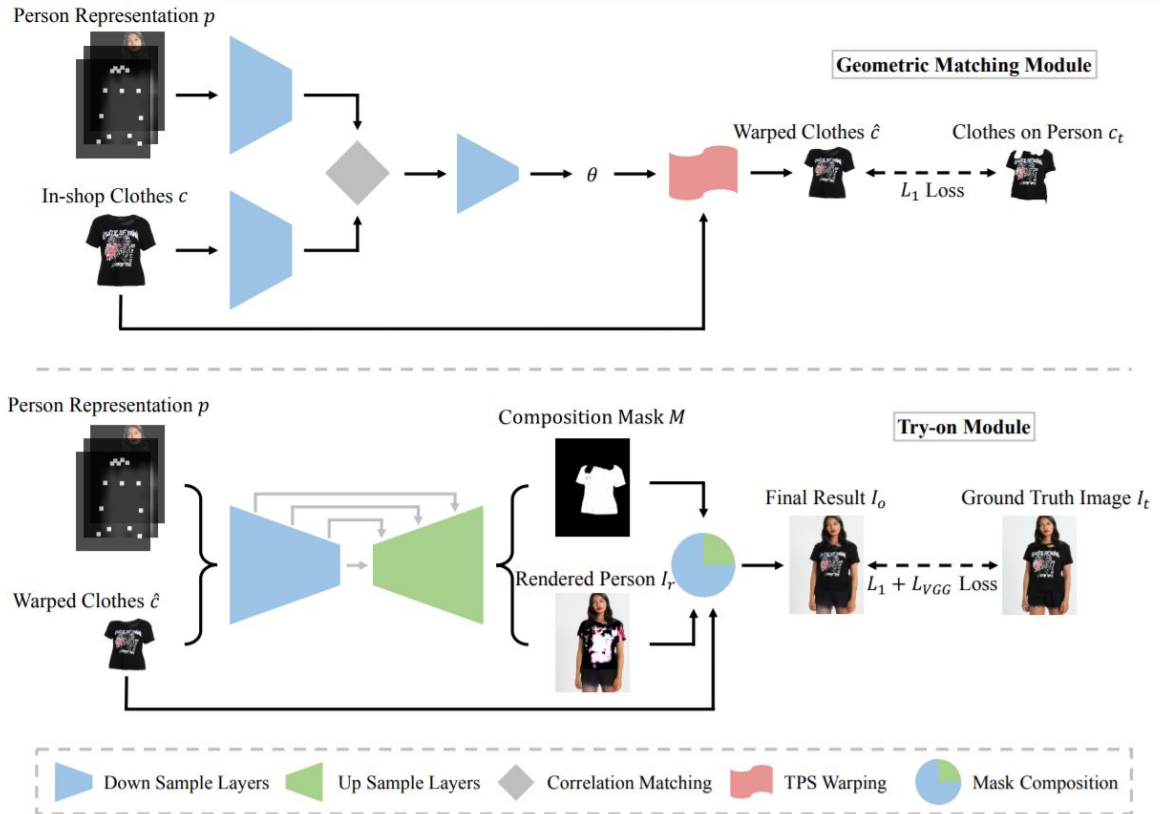
2 - What are Generative Adversarial Networks ?

An impressive propriety of GAN architecture is their capacity to learn without labels. This is called unsupervised learning and it is achieved through two competitives CNN named generator and discriminator. The generative network synthesises, in our case, a fake picture. The discriminative network evaluates the fake picture in comparison to a target set. This evaluation is transmitted gradient to gradient by backpropagation to tell the generator which kernel parameters need to be optimized. This process may seem simple but made the GAN able to learn enough abstraction to recognise a picture as part of a set. GAN develops a deep semantic understanding of what it is like to be part of this set. This semantic called latent space, can be twinkle to generate fake but realistic pictures similar to this set.



CP-VTON is the current state-of-the-art in virtual try-on. For this reason, we will see how its Geometric Matching Module (GMM) and its Try-On Module (TOM) are working.

3 - What are Geometric Matching and Try-On modules ?



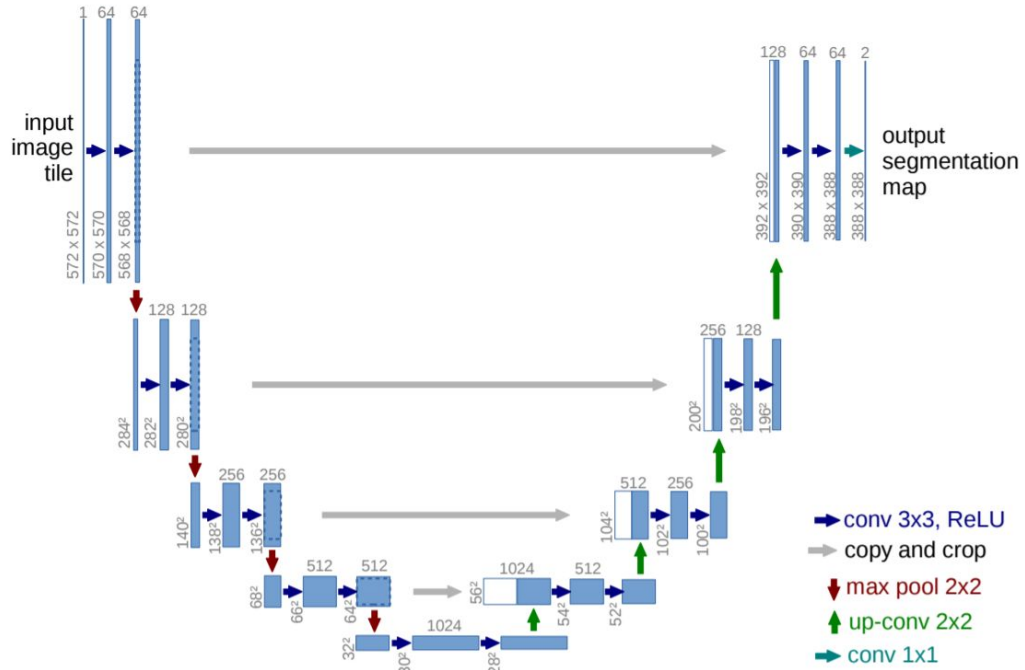
CP-VTON gets triplets p , c and I_t as training inputs. c is the target clothing. I_t is the ground truth cloth on a model image. p is a Person representation used to condition the GAN to generate something close to I_t without giving I_t . p is composed of 3 abstraction of I_t :

- Pose heatmap: an 18-channel feature map with each channel corresponding to one human pose keypoint, drawn as an 11×11 white rectangle.
- Body shape: a 1-channel feature map of a blurred binary mask that roughly covers different parts of the human body.
- Reserved regions: an RGB image that contains the reserved regions to maintain the identity of a person, including face and hair.

They are built with state-of-the-art human parsing such as LIP_JPPNet or CIHP_PGN and a state-of-the-art human pose estimator such as the openpose COCO-18 model.

The GMM is a pipeline with p and c as inputs and c_t , I_t crop clothing, as the ground truth output. Two CNN extract, from p and c , cloth features for correlation matching. A final CNN predicts the spatial transformation θ to apply to c . The loss is the mean absolute error (L_1 Loss) between c_t and $\hat{c}$ warp by θ . A recent promising method, use L_1 loss + Lreg on the mask of $\hat{c}$ and c_t . Lreg regularizes the warping of θ apply

on grid. GMM is trainable on its own, its warp cloths are necessary to train the next stage.



U-Net is an encoder-decoder convolutional network similar to GAN. The second stage of CP-VTON relies on its architecture. To compose the try-on result, the module needs the warp cloth $\hat{c}$, the person image I_r and a composition mask M . UNet takes as inputs the person representation p with $\hat{c}$ and simultaneously synthesise the mask M and a coarse image I_r . The loss function back propagated in the UNet is a combination of the L1 loss and the L_{VGG} loss of the composition result I_o with the ground truth I_t . VGG19 network is a visual perception network (CNN) trained on the classical dataset of real pictures, ImageNet. L_{VGG} is a combination of the L1 distance between feature maps of I_t and I_o , of the first 5 VGG19's convolutional layers. First abstraction levels of VGG19 make L_{VGG} a perceptual loss and guide I_o to realistic looking.

$$\mathcal{L}_{VGG}(I_o, I_t) = \sum_{i=1}^5 \lambda_i \|\phi_i(I_o) - \phi_i(I_t)\|_1$$

PART 2 : What are Virtual Try-On Networks issues ?

1 - Is there implementation and database issues ?

During models re-implementation, a number of bugs and issues were found. Retro-compatibility is a recurrent and major issue in machine learning research. Tensorflow's recent update to v2 forces users to rewrite chunks of VITON and CP-VTON. The human parsing LIP_JPPNet/LIP_SSL update changes the indexation of neck and skin. The head and the neck got their indexation modified, causing Try-on to virtually decapitate the person. CP-VTON+ includes a python correction file to solve this issue. It also provides a python file to generate the body mask using the opencv function. In the future this dependency to other code may continue to cause dysfonctionnement.



Fashion datasets are limited. generation and representability (only women, same light and background). Almost all methode use the same dataset proposed by *Zalando.com*. CAGAN could use its collection of human images paired with their upper-body cloth. Each cloth is on a white background and almost always worn on a frontal view of a woman on a white background in a similar position in regard to human chaos. Later this dataset was increased to 2 poses for each cloth to train FashionTryOn and FashionOn. This means that the model learning is biased by gender, lighting, pose, cloth side, style and origine (even if the set has good variance), perspective and body shape. The set has little cloth direct deformation and not one occlusion outside of the arms and the hair of the model.

Since the database is so selective, algorithms only work for a specific resolution and format. They can handle only one person in the correct position and angle. A code finding the person inside a bigger picture and applying the try-on only to the specific

person frame would make the network much more useful and playful. Strangely, multi person gestion is an issue no paper chooses to discuss.

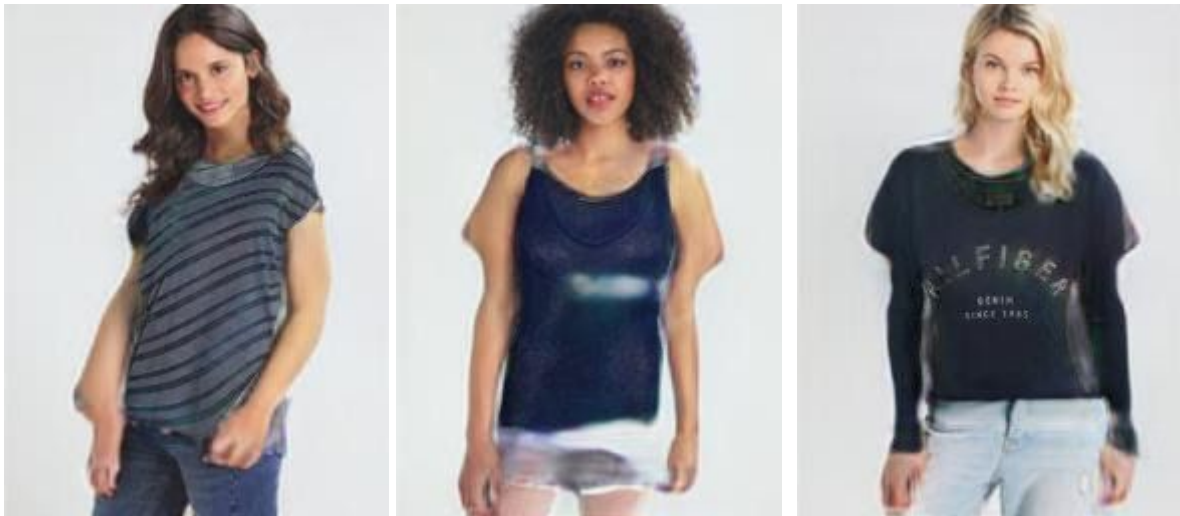
2 - What do failure cases look like ?



CP-VTON+ corrects some minor problems of CP-VTON and gives access to pre-trained models. After some issues and updates, the code generates typical virtual try-on network data. CP-VTON+ has a goodenough architecture to synthesise great results but is not without flowd. Here are somes example of failure cases we can analyse :



The clothes have a neck issue. Their in-shop pictures show the front and the back of the clothing. When the back is not easily detectable it warps and composes onto the result. To move the issue we can train the composition mask to find only the front side or just remove it automatically with the risk of over cropping.



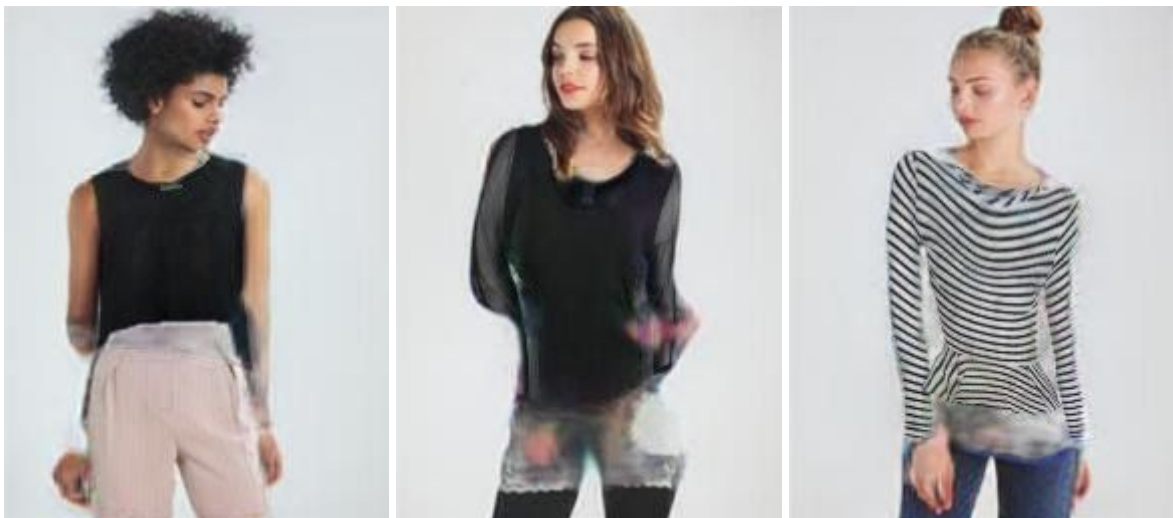
The architecture doesn't allow the initial cloth mask to be filled by the background. When the body does fill the target cloth but not the initial one, gigantic deltoids tend to grow.



The algorithm is trained and optimised for a certain kind of view and frame. Back view are a source of many error.



The warping isn't limited as long as it minimises L1 or VGG loss. This leads to cloth hard deformation.



Since parsing methods are not without failure cases, CP-VTON+ dependence to them is not only a source of arms, hands disappearance and cloth overlapping.



Sometimes there's just nothing working. AI isn't a perfect science.

III - What are the solutions to these issues ?

1 - Are there papers already solving issues ?

Table 11. Summary of the benchmark datasets for style transfer task.

Task	Dataset name	Publish time	# of photos	Key features	Sources
Facial Makeup	Liu <i>et al.</i> [117]	2016	2,000	1000 non-makeup faces and 1000 reference faces	N/A
	Stepwise Makeup [175]	2016	1,275	Images in 3 sub-regions (eye, mouth, and skin); labeled with procedures of makeup	N/A
	Beauty [216]	2017	2,002	1,001 subjects, where each subject has a pair of photos being with and without makeup	The Internet
	Before-After Makeup[3]	2017	1,922	961 different females (224 Caucasian, 187 Asian, 300 African, and 250 Hispanic) where one with clean face and another after professional makeup; annotated with facial attributes	N/A
	Chang <i>et al.</i> [9]	2018	2,192	1,148 non-makeup images and 1,044 makeup images	Youtube makeup tutorial videos
	Makeup Transfer [95]	2018	3,834	1,115 non-makeup images and 2,719 makeup images; assembled with 5 different makeup styles (smoky-eyes, flashy, Retro, Korean, and Japanese makeup styles), varying from subtle to heavy	N/A
	LADN ⁴⁴ [42]	2019	635	333 non-makeup images and 302 makeup images	The Internet
Virtual Try-On	LookBook ⁴⁵ [205]	2016	84,748	The 9,732 top product images are associated with 75,016 fashion model images	Bongjashop, Jogunshop, Stylenanda, SmallMan, WonderPlace
	DeepFashion ⁴⁶ [119]	2016	78,979	The corresponding upper-body images, sentence descriptions, and human body segmentation maps	Forever21
	CAGAN [75]	2017	15,000	Frontal view human images and paired upper-body garments (pullovers and hoodies)	Zalando.com
	VITON ⁴⁷ [51]	2018	32,506	Pairs of frontal-view women and top clothing images	N/A
	FashionTryOn ⁴⁸ [214]	2019	28,714	Pairs of same person with same clothing in 2 different poses and one corresponding clothing image	Zalando.com
	FashionOn [67]	2019	22,566	Pairs of same person with same clothing in 2 different poses and one corresponding clothing image	The Internet, DeepFashion [119]
	Video Virtual Try-on [30]	2019	791 videos	Each video contains 250–300 frame numbers	fashion model catwalk

N/A: there is no reported information to cite.

“Fashion Meets Computer Vision: A Survey” - p19

<https://arxiv.org/pdf/2003.13988.pdf>

This 2019 study tried to find the current best model for diverse fashion related image processing including virtual try-on. It takes only a few papers into account and has not been updated after the release of novel solutions. It is still a useful compilation of similar work and database. Some approaches to the Virtual Try On problem are 3D based. It offers a huge controllability of the final model but does little to actually return a realistic picture. This may help to build 3D models of cloth on people for gaming usage. This approach may be useful to give information on how to deform clothes knowing the physics but is way too slow and resource graving. For this reason they are not discussed in this report.

Here is a non exhaustive list of notable virtual try-on networks :

Traditional virtual try-on papers

CAGAN “The conditional analogy gan: Swapping fashion articles on people images”

N. Jetchev and U. Bergmann, in Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2287–2292.

<https://arxiv.org/pdf/1709.04695.pdf>

Original paper on how to use a GAN with multiple entries to generate a coarse try-on image.

VITON *“Viton: An image based virtual try-on network”*

X. Han, Z. Wu, Z. Wu, R. Yu, and L. S. Davis, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7543–7552.

<https://arxiv.org/pdf/1711.08447.pdf>

The paper that builds the basic architecture around the CAGAN and gave the name “Virtual Try-On” to the art.

CP-VTON *“Toward characteristic-preserving image-based virtual try-on network”*

B. Wang, H. Zheng, X. Liang, Y. Chen, L. Lin, and M. Yang, in Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 589–604.

http://openaccess.thecvf.com/content_ECCV_2018/papers/Bochao_Wang_Toward_Characteristic-Preserving_Image-based_ECCV_2018_paper.pdf

It is the most used, reimplemented and cited one. Full method is detailed upward.

Preserving detail by focusing on the parsing

Those methods only add some fine tuning to the architecture of CP-VTON. They avoid arm removal and pants swapping by focusing on the body segmentation map. It is the parsing and the mask that are used as loss evaluation. The characteristics are preserved through the transformation thanks to this segmentation. SieveNet uses a novel duelling triplet loss strategy to avoid blurry effect on cloth. VTMFP is considered the logical successor of CP-VTON since its architecture is mostly similar and its results are qualitatively and quantitatively better.

VTNFP *“VTNFP: An Image-based Virtual Try-on Network with Body and Clothing Feature Preservation”*

R. Yu, X. Wang, and X. Xie, in 2019 IEEE/CVF International Conference on Computer Vision (ICCV)

http://openaccess.thecvf.com/content_ICCV_2019/papers/Yu_VTNFP_An_Image-Based_Virtual_Try-On_Network_With_Body_and_Clothing_ICCV_2019_paper.pdf

CP-VTON+ *“CP-VTON+: Clothing Shape and Texture Preserving Image-Based Virtual Try-On”*

https://minar09.github.io/cpvtonplus/cvprw20_cpvtonplus.pdf

“Image-based virtual try-on network with structural coherence”

F. Sun, J. Guo, Z. Su, and C. Gao, in 2019 IEEE International Conference on Image Processing (ICIP). IEEE, 2019, pp. 519–523.

<https://sci-hub.tw/downloads/2019-11-04/31/sun2019.pdf?rand=5f440c937ae06?download=true>

“Powering Virtual Try-On via Auxiliary Human Segmentation Learning”

K. Ayush, S. Jandial, A. Chopra, and B. Krishnamurthy, in 2019 ICCVW.

https://openaccess.thecvf.com/content_ICCVW_2019/papers/CVFAD/Ayush_Powering_Virtual_Try-On_via_Auxiliary_Human_Segmentation_Learning_ICCVW_2019_paper.pdf

SieveNet *“SieveNet: A Unified Framework for Robust Image-Based Virtual Try-On”*

<https://arxiv.org/pdf/2001.06265.pdf>

FashionOn *“Fashionon: Semantic-guided image-based virtual try-on with detailed human and clothing information”*

C.-W. Hsieh, C.-Y. Chen, C.-L. Chou, H.-H. Shuai, J. Liu, and W.-H. Cheng, in Proceedings of the 27th ACM International Conference on Multimedia. ACM, 2019, pp. 275–283.

<https://dl.acm.org/doi/pdf/10.1145/3343031.3351075>

http://39.96.165.147/Pub%20Files/2019/hcw_mm19.pdf

FashionOn leaves the traditional coarse mask to focus on the parsing. The goal is to synthesise a good parsing of the result. It is done by a first conditional GAN that inputs the parsing of the model, the target pose and the target cloth mask. The segmentation map output is then colorise by a second conditional GAN thanks to the target cloth and the model without cloth. FashionOn parsing approaches made possible a refinement specialise for the face, with FacialGAN, and for the cloth, with Clothing UNet. The results may be cherry-picked but are convincing non the less.

Solving cloth distortion

WUTON *“Do Not Mask What You Do Not Need to Mask: a Parser-Free Virtual Try-On”*

<https://arxiv.org/pdf/2007.02721.pdf>

Since parsing methods are not without failure cases, CP-VTON dependence to them is not only a source of failure but also a time and computing loss. This method builds a student try-on network (S-WUTON) that learns without parsing from a teacher similar to CP-VTON (T-WUTON). S-WUTON is then a parser free architecture

avoiding previous issues. Its runtime (13 ms) is less than a tenth of CP-VTON, ClothFlow or VTNFP ones (around 175 ms).

LA-VITON *“LA-VITON: A Network for Looking-Attractive Virtual Try-On”*

H. J. Lee, R. Lee, M. Kang, M. Cho, and G. Park, in 2019 ICCVW.

https://openaccess.thecvf.com/content_ICCVW_2019/papers/CVFAD/Lee_LA-VITON_A_Network_for_Looking-Attractive_Virtual_Try-On_ICCVW_2019_paper.pdf

Two of the major issues of CP-VTON are elegantly solved by LA-VITON. The warping wasn't restricted, this leads to distortion in the clothing. By warping in two steps, perspective and Thin plate spline (TPS) for data interpolation and smoothing. Grid Interval Consistency loss (GIC) is measured on the TPS warping. Combining this GIC in the GMM loss tends to limit the distortion of the cloth after perspective. The result is texture and motif consistency and realism. The GMM loss combination is design and tune to handle occlusion. This means that the cloth mask is not restricted by the arm or hair in the frame. This occlusion handling and GIC made LA-VITON one of the best virtual try-on methods.

ACGPN *“Towards Photo-Realistic Virtual Try-On by Adaptively Generating↔Preserving Image Content”*

<https://arxiv.org/pdf/2003.05863.pdf>

ACGPN is similar to LA-VITON in its grid warping constraint. The data are sorted in three complexity categories from easy to hard. Its new architecture and results makes it one of the new possible state-of-the-art compared to CP-VTON or VTNFP.

Multi-pose handling

Those papers had the idea to combine the virtual try-on with a pose transformation. This allows the user to see himself wearing the cloth in any given position. VTNCAP relies on an Attentive Bidirectional GANs that render the expected result then render an image with the original pose and cloth. Fit-me changes the Try-On Module into a rough and precise pose transformation. They are conditional GANs specialise on respectively pose approximation and details refining. MG-VTON being open source, it is the one currently used for comparison. Clothflow base its pose-guided transformation on flow warping to preserve most of the texture and detail.

Fit-me *“Fit-me: Image-based virtual try-on with arbitrary poses”*

C.-W. Hsieh, C.-Y. Chen, C.-L. Chou, H.-H. Shuai, and W.-H. Cheng, in 2019 IEEE International Conference on Image Processing (ICIP). IEEE, 2019, pp. 4694–4698.

<https://dacemirror.sci-hub.st/proceedings-article/f30e56f6cfec394cdb9aa4b879f34d36/hsieh2019.pdf?rand=5f4149e0d8a74?download=true>

MG-VTON, “Towards Multi-pose Guided Virtual Try-on Network”

H. Dong, X. Liang, X. Shen, B. Wang, H. Lai, J. Zhu, Z. Hu, and J. Yin, in 2019 ICCV.

https://openaccess.thecvf.com/content_ICCV_2019/papers/Dong_Towards_Multi-Pose_Guided_Virtual_Try-On_Network_ICCV_2019_paper.pdf

ClothFlow “ClothFlow: A Flow-Based Model for Clothed Person Generation”

http://openaccess.thecvf.com/content_ICCV_2019/papers/Han_ClothFlow_A_Flow-Based_Model_for_Clothed_Person_Generation_ICCV_2019_paper.pdf

VTNCAP “Virtually trying on new clothing with arbitrary poses”

N. Zheng, X. Song, Z. Chen, L. Hu, D. Cao, and L. Nie, in 2019 Proceedings of the 27th ACM International Conference on Multimedia. ACM, pp. 266–274.

<https://xuemensong.github.io/fp452-zhengA.pdf>

“Down to the Last Detail : Virtual Try-On with Detail Carving”

<https://arxiv.org/pdf/1912.06324.pdf>

“Toward Accurate and Realistic Virtual Try-on Through Shape Matching and Multiple Warps”

<https://arxiv.org/pdf/2003.10817.pdf>

This last one uses a shape matching algorithm to choose the right details to carry and a warping model that learns multiple coordinated-warps.

Other paper more general than virtual try-on

FashionGAN “Be your own Prada: Fashion synthesis with structural coherence”

S. Zhu, R. Urtasun, S. Fidler, D. Lin, and C. C. Loy, in 2017 ICCV.

<https://arxiv.org/pdf/1710.07346.pdf>

M2e-TON “M2e-try on net: Fashion from model to everyone”

Z. Wu, G. Lin, Q. Tao, and J. Cai, in 2019 Proceedings of the 27th ACM International Conference on Multimedia. ACM, pp. 293–301.

<https://arxiv.org/pdf/1811.08599.pdf>

The originality of those two papers is the input form. FashionGAN generates clothes based on a text description. M2e-TON takes the target clothing from another wearer image than an in-shop one. Their results are difficult to evaluate and to compare but looks promising.

LGVTON “LGVTON: A Landmark Guided Approach to Virtual Try-On”

<https://arxiv.org/pdf/2004.00562v1.pdf>

LGVTON improved on M2e-TON by introducing a new heatmap optimise for clothing. This optimises the matching and the warping.



SMIS “Semantically Multi-modal Image Synthesis”

<https://arxiv.org/pdf/2003.12697.pdf>

This architecture is very different from the other. SMIS builds a latent space to synthesise realistic cloth on a person's picture. This latent space made the morphing from one cloth to another easy and realistic at every stage. It gives a great controllability to the final result aspect and clothing.

FW-GAN “FW-GAN: Flow-Navigated Warping GAN for Video Virtual Try-On”

H. Dong, X. Liang, X. Shen, B. Wu, B. Chen, and J. Yin, in 2019 ICCV.

https://openaccess.thecvf.com/content_ICCV_2019/papers/Dong_FW-GAN_Flow-Navigated_Warping_GAN_for_Video_Virtual_Try-On_ICCV_2019_paper.pdf

Current methods like CP-VTON are made for photography without continuity. This means that a video virtual try-on will have flickering clothes and a lot of artefacts. FW-GAN takes into account the previous frames and poses sequences to render a coherent and realistic try-on.

2 - How to measure a good try-on ?

To evaluate the quality of a virtual try-on, studies have asked the user. On the Amazon Mechanical Turk (**AMT**) platform repetitive tasks like those can be asked to an arbitrary number of people. If it is a useful measurement to compare several results and a ground truth before publishing, this is resource and time consuming. To get a quantitative measure fast and reproducible a all set of formula are disponible :



Cross-entropy loss, or log loss, measures the performance of a classification model whose output is a probability value between 0 and 1. Cross-entropy loss increases as the predicted probability diverges from the actual label.

Mean Absolute Error (**MAE**), or **L1** loss is the absolute distance between the synthesized image and the ground truth. Some papers choose to compare the mask of those images. Mean Squared Error (**MSE**), or **L2** loss is the same but the distance is squared.

Peak signal-to-noise ratio (**PSNR**) is the ratio between the maximum possible pixel value of an image and the power of corrupting noise that affects the fidelity of its representation. In dB, it is defined as $10 \cdot \log_{10}(\text{MAX}_i^2 / \text{MSE})$.

VGG networks are visual perception networks (CNN) trained on the classical dataset of real pictures, ImageNet. Firsts layers gives features maps detects small details and texture. The VGG loss is a combination of the L1 loss of those features maps. Similar losses are constructed from other CNN.

The Inception score (**IS**) is a popular metric for judging the image outputs of GANs. This score is based on a statistics formula called the Kullback-Leibler (**KL**) divergence. The KL divergence is a measure of how similar/different two probability distributions are. IS favor realistic results and variety in the set.

The structural similarity index measure (**SSIM**) is a perception-based model that considers image degradation as perceived change in structural information, while also incorporating important perceptual phenomena. Its formula is based on three comparison measurements between the samples two images : luminance, contrast and structure. Performing better on certain sets, Multi-Scale SSIM (**MS-SSIM**) is conducted over multiple scales through a process of multiple stages of sub-sampling, reminiscent of multiscale processing in the early vision system.

Learned Perceptual Image Patch Similarity (**LPIPS**) metric. Evaluate the distance between image patches. “We found that deep network activations work surprisingly well as a perceptual similarity metric. This was true across network architectures (SqueezeNet [2.8 MB], AlexNet [9.1 MB], and VGG [58.9 MB] provided similar scores) and supervisory signals (unsupervised, self-supervised, and supervised all perform strongly). We slightly improved scores by linearly "calibrating" networks - adding a linear layer on top of off-the-shelf classification networks. We provide 3 variants, using linear layers on top of the SqueezeNet, AlexNet (default), and VGG networks.” Ensembled LIPS (**E-LIPS**) apply a random deformation in 4 steps before calculating the PIPS loss.

Grid interval consistency (**GIC**) loss retains the characteristics of clothes after warping. TPS transformation generally shows good performance but its high flexibility often causes distortion of patterns and prints. Even when the coarse-to-fine strategy is applied, TPS with a high degree of freedom causes undesirable warping results. Clothes are deformable objects, but deformation on a human torso is very limited. Therefore, geometric matching needs to be restricted to maintain the shape and appearance of clothing.

Frechet Inception Distance (**FID**) is a metric for evaluating image synthesis quality. It uses an inception model as a feature extractor after removing the last few layers of the network, and extracts feature vectors from real images and synthesized images respectively.